

Factorial one-sided count certificates on the odd reciprocal ES spectrum

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Abstract

The exact denominator defects of the Erdős–Straus divisor atlas are odd. Explicit rational interpolation polynomials therefore separate successes from failures with factorial uniform error and only linear growth of the absolute coefficient sum. Preserving the original state counts gives simultaneous one-sided certificates for every canonical row defined here, for primes $p \equiv 1 \pmod{4}$, at moment degree asymptotic to $\log p / \log \log p$. The finite coefficient-error and split-compression maps are explicit. The theorem controls recovery of the count; it does not prove that every such prime's count is positive. No historical-priority claim is made.

Exact arithmetic domain. For prime $p = 4m + 1$, retain every $m + 1 \leq a \leq 2m$, $R = 4a - p$, $a = \prod_i q_i^{e_i}$, $-e_i \leq \beta_i \leq e_i$, $u = \prod_i q_i^{e_i + \beta_i}$, and the two separate channel tags E, M . Their gates and defects are

$$g_E = 4u + 1, \quad g_M = u + a, \quad D = R / \gcd(R, g_c).$$

The original positive rational triples are

$$E : \left(a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \quad M : \left(a, \frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right).$$

Multiplication gives $(RY - pa)(RZ - pa) = p^2a^2$, proving their reciprocal sum is $4/p$. Since $\gcd(pa, R) = 1$ and $4a \equiv p \pmod{R}$, both nonfirst reduced denominators are exactly D . Thus the hit indicator is $\mathbf{1}_{D=1}$, and every failure has odd $D \geq 3$. The inherited complete first-half atlas retains the Type I/II coordinate conventions [2, 3]. The original inverse is $u = a^2/(RY - pa)$ for E and $u = pa^2/(RY - pa)$ for M, with its tag.

For the countable disjoint union of these labelled rows use counting ℓ^2 , $Xe_\alpha = D_\alpha^{-1}e_\alpha$, and $\Pi e_\alpha = \mathbf{1}_{D_\alpha=1}e_\alpha$. The spectral accumulation point zero is not an absent state. The exact ambient compact set is

$$\Sigma = \{0, 1, 1/3, 1/5, 1/7, \dots\}.$$

Theorem 1 (Discrete signed interpolation). *Define the degree- $(j + 1)$ polynomial*

$$\Phi_j(x) = x \prod_{\nu=1}^j \frac{(2\nu + 1)x - 1}{2\nu}, \quad j \geq 0. \quad (1)$$

Its value is one at 1. At $D = 2k + 1 > 1$ its value is zero for $1 \leq k \leq j$; otherwise it is

$$\Phi_j(1/D) = (-1)^j \frac{\binom{k-1}{j}}{(2k+1)^{j+1}}. \quad (2)$$

Consequently on Σ ,

$$\Phi_{2r+1} \leq \mathbf{1}_{\{1\}} \leq \Phi_{2r}, \quad \|\Phi_j - \mathbf{1}_{\{1\}}\|_\Sigma \leq \frac{1}{2^j j! (2j + 3)}. \quad (3)$$

The even sequence decreases and the odd sequence increases to the indicator. The coefficient ℓ^1 norm of Φ_j is exactly $j + 1$.

Proof. Each factor of (1) is $(\nu - k)/(\nu D)$ at $1/D$, giving (2). For $k > j$, its absolute value is at most $1/(2^j j! D)$, and $D \geq 2j + 3$. The two additional factors in the same-parity comparison have product in $[0, 1]$ on each remaining failed node; newly reached nodes become zero. This proves the signs and monotonicity. All nonzero coefficients alternate; their absolute sum is $|\Phi_j(-1)| = \prod_{\nu=1}^j (\nu + 1)/\nu = j + 1$. $\square$

Interpolation and exact error. The full evaluation map on $0, 1, 1/3, \dots, 1/(2j+1)$ is a rational linear isomorphism. Its inverse is the Lagrange formula

$$(v_i)_i \mapsto \sum_i v_i \prod_{h \neq i} \frac{x - z_h}{z_i - z_h}.$$

Thus (1) is one explicitly selected inverse column of a classical map [1, §3.3(i), equations 3.3.1–3.3.3]. Its values outside these nodes are not asserted to have the signs in (3); the discrete spectrum is essential.

There is a sharp finite alternative to the bound in (3). For $j \geq 1$ the maximum in (2) occurs among the integers

$$\max\{j+1, \lfloor 1 + j(j+2)/2 \rfloor\} \leq k \leq \max\{j+1, \lceil j + j(j+2)/2 \rceil\}. \quad (4)$$

Indeed the logarithmic derivative of the positive expression in (2) has the sign of $\sum_{\nu=1}^j (2\nu+1)/(k-\nu) - 2$, a strictly decreasing function. Its crossing lies between $1 + j(j+2)/2$ and $j + j(j+2)/2$ by bounding every denominator between $k-j$ and $k-1$. This proves (4), including the endpoint restriction. Exact evaluation gives errors $1/3, 2/49, 10/2197, 4/9261$ at $j = 0, 1, 2, 3$, attained respectively at $D = 3, 7, 13, 21$.

Original mass, not a probability replacement. Let W_p be the size of a full canonical row and H_p its success count. An elementary bound is

$$W_p \leq B_p = \min\{6m^2, 4m(1 + \lfloor \log_2(2m) \rfloor)^2\}. \quad (5)$$

For the first expression use $\tau(a^2) \leq 2a-1$. For the second use the prime-power inequality $2e+1 \leq (e+1)(e+2)/2$, hence $\tau(a^2) \leq d_3(a)$. Then $\sum_{a \leq N} d_3(a) \leq N(\sum_{u \leq N} 1/u)^2$, and dyadic grouping bounds the harmonic sum by $1 + \lfloor \log_2 N \rfloor$. Double for both channels.

Theorem 2 (Simultaneous original-count certificates). *For accuracy index $n \geq 0$, choose the least even $j = j(p, n)$ with*

$$2^j j! (2j+3) \geq 8 \cdot 3^n B_p.$$

Set $Q_j(p) = \sum_{\alpha \in \mathcal{A}_p} \Phi_j(D_\alpha^{-1})$. Then

$$\boxed{H_p - \frac{1}{16 \cdot 3^n} \leq Q_{j+1}(p) \leq H_p \leq Q_j(p) \leq H_p + \frac{1}{8 \cdot 3^n}.} \quad (6)$$

In particular $\lceil Q_{j+1}(p) \rceil = \lfloor Q_j(p) \rfloor = H_p$, and $H_p > 0$ if and only if $Q_{j+1}(p) > 0$. The largest required moment index is $j+2 = (1 + o(1)) \log p / \log \log p$ for fixed n .

Proof. Sum (3) with the original unit masses and apply (5). The ratio between consecutive error denominators is $2(j+1)(2j+5)/(2j+3) > 2$. This proves (6) and its rounding consequence. The estimate $\log(2^j j! (2j+3)) = j \log j + O(j)$ follows by integrating $\log x$ above and below its integer sum. Minimality among even integers changes it by $O(\log j)$; $\log B_p = \log p + O(\log \log p)$ proves the degree. $\square$

The inequality $Q_{j+1}(p) > 0$ is not established for every prime by this convergence theorem. It is the now-explicit arithmetic lower bound to prove. On the original prime counting space, the number operator $N = \text{diag } H_p$ retains its maximal domain $\sum H_p^2 |v_p|^2 < \infty$. The two operators in (6) have that same domain and bounded differences from N with the displayed uniform constants.

Certified moment inputs. Write $M_k(p) = \sum_{\alpha} D_{\alpha}^{-k}$. Each Q_j is an explicit rational linear combination of these original moments. If every moment through degree $j + 2$ has certified absolute error at most

$$\delta = \frac{1}{16(j+2)3^n},$$

the total evaluation error is at most $1/(16 \cdot 3^n)$ for either filter. Form the lower endpoint by subtracting this allowance and the upper by adding it to the approximate sums. Their width is at most $7/(16 \cdot 3^n) < 1$ and they contain the integer H_p . This is an absolute error statement. It does not claim that computing the original moments to this accuracy has low complexity.

For any finite original selection, the particularly simple quartic gives

$$\Phi_3(x) = \frac{105x^4 - 71x^3 + 15x^2 - x}{48} \leq \mathbf{1}_{\{1\}}(x).$$

At one shell put $S_k = \sum_{\alpha} \gcd(R, g_{\alpha})^k$. The unconditional sufficient test is therefore

$$\boxed{105S_4 - 71RS_3 + 15R^2S_2 - R^3S_1 > 0 \implies H > 0.} \quad (7)$$

A separate accompanying note supplies a hard-prime shell at which (7) succeeds while the optimal aggregate first-degree signed moment test fails. Neither (7) nor first-degree adequacy is asserted necessary for arbitrary positive shells.

Split-fibre and operator return. For $e = f + b$ the addition map $(\xi, \eta, c) \mapsto (\xi + \eta, c)$ has complete fibre

$$\max(-f_i, \beta_i - b_i) \leq \xi_i \leq \min(f_i, \beta_i + b_i), \quad \eta = \beta - \xi.$$

Its size $n(\beta)$ is the product of those interval lengths. Assigning each occurrence weight $1/n(\beta)$ preserves every M_k and Q_j exactly by finite fibre summation. All channel labels and original exponent vectors remain; a residue-support subgroup does not replace the exponent box.

In the completed occurrence space with those weights, value duplication U has adjoint equal to fibre averaging, and $U^*U = I$. For $A = UXU^*$ on the full occurrence space, every polynomial satisfies

$$f(A) = Uf(X)U^* + f(0)(I - UU^*).$$

Decompose into $\text{im } U$ and its orthogonal complement to prove it. Because $\Phi_j(0) = 0$,

$$\boxed{\Phi_j(UXU^*) = U\Phi_j(X)U^*} \quad (8)$$

The return on the image uses U^* ; the kernel directions stay marked. The raw repeated diagonal trace is different and multiplies original states by their fibre sizes. For the product in (1) without the leading x , its constant term would instead contribute $(-1)^j(2^j j!)^{-1}(I - UU^*)$.

Verification and scope. The accompanying standard-library verifier uses rational arithmetic and checks all signs, coefficient sums, interpolation inverse columns, sharp finite windows, original split fibres and the stated operator comparisons. It checks the row schedule through $p \leq 1500$ and on explicit growing-mass fixtures. The infinite theorems are proved above, not inferred from that finite replay. Original normalized ES coordinates recover a witness after a positive sum identifies a successful finite source label. Moments alone do not identify which state supplied it. Supported coefficient zero is not external absence.

References

- [1] National Institute of Standards and Technology. *Digital Library of Mathematical Functions*, §3.3(i), equations 3.3.1–3.3.3.
- [2] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *J. Aust. Math. Soc.*, 94(1), 50–105, §2, Propositions 2.2 and 2.6.
- [3] The Clankers (2026). *Uniform infinite-array control for marked Erdős–Straus selectors*, 14 September, Lemma 1 and Theorem 2. Complete source/hash identity and the preceding SplitZero integration are in the companion ledger.